CL117-INTRODUTION TO INFORMATION AND COMMUNICATION TECHNOLOGIES

Lab #: 5

INTRODUCTION TO MS EXCEL

SCHOOL OF COMPUTING FAST-NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES, ISLAMABAD CAMPUS

I- Introduction to MS Excel

Launch Excel

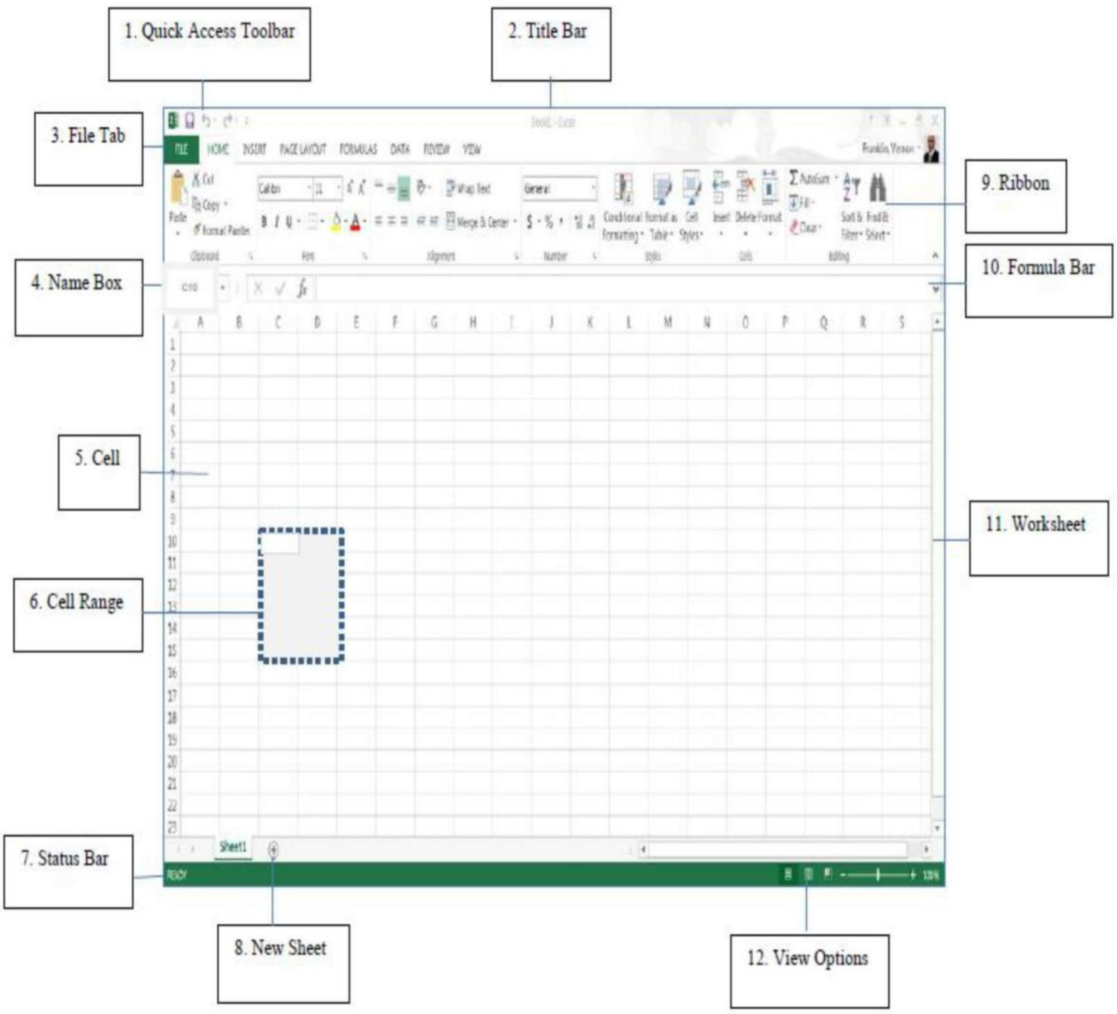
To launch Excel for the first time:

1. Click on the Start button.
2. Click on All Programs.
3. Select Microsoft Office from the menu options, and then click on Microsoft Excel.

Note: After Excel has been launched for the first time, the Excel icon will be located on the Quick Launch pane. This enables you to click on the Start button, and then click on the Excel icon to launch the Excel spreadsheet. Also, a shortcut for Excel can be created on your desktop.

Window Features

The purpose of the window features is to enable the user to perform routine tasks related to the Microsoft applications. All the Office applications share a common appearance and similar features. The window features provide a quick means to execute commands. Here are some pertinent Excel features:

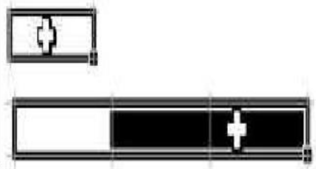
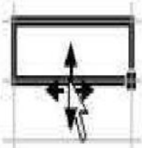


Spreadsheet Terms

	Term	Description
1	Quick Access Toolbar	Displays quick access to commonly used commands.
2	Title Bar	Displays the name of the application file.
3	File Tab	The File tab has replaced the Office button. It helps you to manage the Microsoft application and provide access to its options such as Open, New, Save, As Print, etc.
4	Name Box	Displays the active cell location.
5	Cell	The intersection of a row and column; cells are always named with the column letter followed by the row number (e.g. A1 and AB209); cells my contain text, numbers and formulas.
6	Range	One or more adjacent cells. A range is identified by its first and last cell address, separated by a colon. Example ranges are B5:B8, A1:B1 and A1:G240.
7	Status Bar	Displays information about the current worksheet.
8	New Sheet	Add a new sheet button.
9	Ribbon	Displays groups of related commands within tabs. Each tab provides buttons for commands.
10	Formula Bar	Input formulas and perform calculations.
11	Worksheet	A grid of cells that are more than 16,000 columns wide (A-Z, AA-AZ, BA-BZ…IV) and more than 1,000,000 rows long.
12	View Option	Display worksheet view mode.

Mouse Pointer Styles

The Excel mouse pointer takes on many different appearances as you move around the spreadsheet. The following table summarizes the most common mouse pointer appearances:

Pointer	Example	Description
		The white plus sign will select a single cell to enter data, retype data or delete text from the selected cell. This pointer is also useful for selecting a range of cells.
		The white arrow will drag the contents of the selected cell to a new location (drag and drop).
		The black plus sign activates the fill handle of the selected cell and will fill the adjoining cells with some type of series, depending on the type of data (e.g., a formula or date) is in the beginning cell.

Spreadsheet Navigation

The following table provides various methods to navigation around a spreadsheet.

Method	Description
mouse pointer	Use the mouse pointer to select a cell.
scroll bars	Use the horizontal and vertical scroll bars to move around the spreadsheet to view columns and rows not currently visible. Click the mouse pointer once the desired cell is visible.
arrow keys	Use the left ←, right →, up ↑, and down ↓ arrows to move accordingly among cells.
Enter	Press the Enter key to move down one cell at a time.
Tab	Press the Tab key to move one cell to the right.
Ctrl+Home	Moves the cursor to cell A1.
Ctrl+End	Moves the cursor to the last cell of used space on the worksheet, which is the cell at the intersection of the right-most used column and the bottom-most used row (in the lower-right corner).
End + arrow key	Moves the cursor to the next or last cell in the current column or row which contains information.

1. Practice moving around the spreadsheet.
2. Practice selecting cells and cell ranges.

Basic Steps for Creating a Spreadsheet

When creating a spreadsheet it is recommended to do the following steps:

1. Made a draft of your spreadsheet idea on paper.
2. Enter the data from your draft onto the actual spreadsheet.
3. Format your data after entering onto the spreadsheet.
4. Calculate data by using mathematical formulas.
5. Save the document.
6. Preview and Print the spreadsheet.

II- Enter and Format Data

Create Spreadsheet

1. Illustration of spreadsheet to be completed in exercise below:

Budget for Guest Speakers				
Item	**Fall**	**Spring**	**Summer**	**Annual**
Research	20	20	10	50
Correspondence/Communication	30	30	15	75
Publicity	50	50	25	125
Honorariums	500	500	250	1250
Travel	750	750	325	1825
Lodging	300	300	150	750
Total	$1,650.00	$1,650.00	$ 775.00	$4,075.00

2. Open **Excel Practice File.xlsx** , and then click on the **Budget sheet**


tab.
(The instructor will indicate the location for this file.)

   a. Select cell **A1**, and then type **Budget for Guest Speakers.**
   b. Select cell **A3**, type **Item**, and then press the **Tab** key.
   c. Select cell **B3**, type **Fall**, and then press the **Tab** key.
   d. Select cell **C3**, type **Spring**, and then press the **Tab** key.
   e. Select cell **D3**, type **Summer**, and then press the **Tab** key.
   f. Select cell **E3**, type **Annual**, and then press the **Tab** key.

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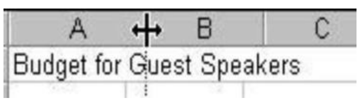
Adjust Column Width

Initially all columns have the same width in a spreadshee . Often you will need to make columns wider or narrower. For example, a long text entry in one cell will be cut off/truncated when the cell to its right contains any information. Likewise, numbers will appear as pound symbols ### when larger than cell width. There are several ways to modify column width.

method	**Description**
dragging method 	Move the cursor up to the column heading area and point to the vertical line to the right of the column that you want to change. When the cursor becomes a "plus sign" with horizontal arrows, press the mouse button and drag in either direction to resize the column. Release the mouse button to accept the new size.
double click to auto fit	Move the cursor up to the column heading area and point to the vertical line to the right of the column that you want to change. When the cursor becomes a "plus sign" with horizontal arrows, double click to AutoFit this one column.
AutoFit a range	Use the mouse to select the range of cells that needs to be adjusted and on the **Home** ribbon in the **Cells** group, choose **Format**, and the select the **AutoFit Column Width** option.

1. **Increase** the width of **column A** via the dragging method so that all text entries are visible.

2. **Decrease** the width of **column C** via the dragging method until pound symbols ### appear.

3. **Increase** the width of **column C** to return to its original size.

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Type Text and Numbers

Use the plus sign mouse pointer to select a cell then begin typing in that cell to e ter data. If there is existing text/data in a cell, the new text will replace the existing text. Press the **Enter** or **Tab** key after typing text in a cell.

1. Type the following text and numbers in rows 10 and 11:

10	Food	90	90	45
11	Total			

Undo and Redo

Use the **Undo** button to undo (reverse) previous actions in reverse sequence. Choose this option immediately after performing an unwanted action. Note that Undo is not available for all commands. The **Redo** button will restore the process that was just undone.

1. Click on the **Undo** button. The last item that you typed is removed from the spreadsheet.

2. Click on the **Redo** button. The text that you removed with Undo should be replaced.

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Insert nd Delete Rows and Columns

Insert rows and columns to add information between exis ing rows or columns of information.

Procedure	Description
Add Row	Select any cell of the row where you desire to add a new row above. On the **Home** ribbon in the Cell group, click on the **Insert** button, and then select **Insert Sheet Rows**. A new roll will appear above your selected cell row.
Add Column	Select any cell of the column letter where you desire to add a new column to the left. On the **Home** ribbon in the Cell **group**, click on the **Insert** button, and then select **Insert Sheet Columns**. A new column will appear to the left of your selected column.
Delete Row or Column	Select any cell where you desire to delete a row or column. On the **Home** ribbon in the **Cell** group, click on the **Delete** button, and then selected **Delete Sheet Rows** or **Delete Sheet Columns**. The row or column where the cell was selected will be deleted.

1. Select any cell in **column C**.
2. On the **Home** ribbon in the **Cell** group, click on the **Insert** drop-down arrow, and then select **Insert Sheet Columns**. A new column will appear to the left of your selected column.
3. Click the **Undo**  button.
4. Select any cell in **row 6**.
5. On the **Home** ribbon in the **Cell** group, click on the **Insert** drop-down arrow, and then select **Insert Sheet Rows**. A new roll will appear above your selected cell row.
6. Select cell **A6,** and then type **Photocopy Services**.
7. Press **Tab** and complete the additional columns as follows:

B	C	D
75	65	30

a

l n

Text nd Number Alignment

Microsoft Excel aligns data in a cell in three ways; left, center, and right. A so, a ra ge of cells can be merged into one cell; this is good for text titles. The default text alignment is left, and the default number alignment is right. Alignment can be changed by using the alignment icons located on the **Home** ribbon in the **Paragraph** group.
Select a range before changing alignment to more than one cell at a time.

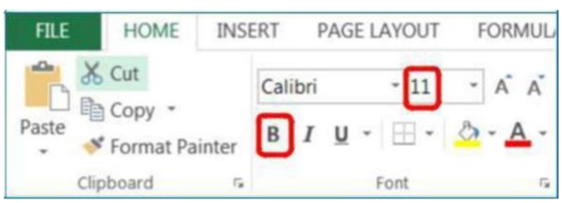
1. Select cell **A3,** and then click on the **Center** alignment button, located on the **Home** ribbon.

2. Select the range **B3:E3**, and the click on the **Center** alignment button, located on the **Home** ribbon.

3. Select the range **A1:E1,** and then click on the **Merge & Center** button, located on the **Home** ribbon.

Format Fonts

Character formats include changing the font, point size, and style of text or numbers.
The fastest way to change fonts is to use the associated buttons on the **Home** ribbon:



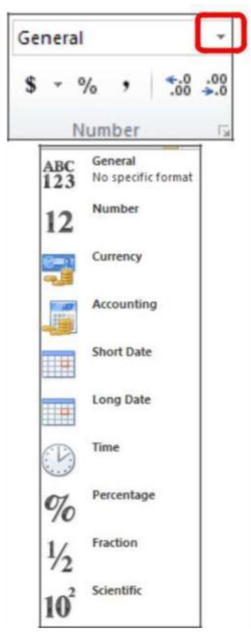
1. Select cell **A1**, and Increase the point size for the title, by clicking on the drop-down arrow on the **Font size** button.

2. Select cell range **A3:E3**, and then click on the **Bold** button to bold text.

Note: To select all cells on a worksheet, click the gray rectangle in the upper-left corner of a worksheet where the row and column headings meet.
Once you select the worksheet, any format change you make will affect the entire sheet.

Form t Numbers

Excel provides many different types of numeric forma s including currency, perce t, comma, scientific, etc. On the **Home** ribbon the numeric formats are located in the **Number** group. Select the drop-down arrow next to **General** to view all format types. Select a range of cell/s before choosing format. In fact, this range can include cell/s that does not yet contain data.



1. Select the cell range **B4:E12**.

2. Click on the **Comma** button, located on the **Home** ribbon.

3. With that same range selected, click on the **Currency** button, located on the **Home** ribbon.

4. To display only dollars and no cents, click on the **Decrease Decimal** button, located on the **Home** ribbon.

Note: To remove a number format from cells, select the **General** format option from the Number group.

Cut, Copy, and Paste Text

Avoid retyping in Excel by moving or copying text and formulas. The following list i cludes commands and definitions involved in cut, copy, and paste.

Command	Description
Cut 	Removes the selected text from the document and places it in the clipboard (a temporary holding place for the item that has been cut or copied).
Copy 	Places a copy of the selected text in the clipboard and leaves the selected text unchanged.
Paste 	Places text from the clipboard in the document where the active cell is located.

Suppose you want to show an identical budget for an additional year. In this exercise, you will copy data in cell range A3:E13, then paste it to sheet2.

1. Select cell range **A3:E13**.
2. Click on the **Copy** button, located on the **Home** ribbon.
3. Click on the **Copy sheet**  tab.
4. **Select** cell **A3**, and then click on the **Paste** button, located on the **Home** ribbon.
5. Click on the **Undo**  button to clear data from spreadsheet. This sheet will be used again for another exercise.

Note: When you copy a range, a moving border with appear around the selected area. Once you paste the data to remove the moving border, double click in any cell outside of the selected range.

Print a Spreadsheet

Click on the **File** tab, and select the **Print** option. Preview your spreadsheet on the right-hand side of the File screen. If you are satisfied with the preview, click the **Print** button, otherwise click on the Home tab to return to the document and edited document.

Exit Excel

When you are finished using Excel, use click on the **File** tab, and select the **Exit** option or click on the Close  button in the upper right-hand corner of the Excel window. If your file has recently been saved, Excel will exit promptly. However, if the file needs to be saved before quitting, Excel will prompt you to save.

III. Basic Formulas

Microsoft Excel is an electronic spreadsheet that automates manual calcu ations involved in accounting and bookkeeping. After you have typed the basic text and number entries in a spreadsheet cell, Excel can perform the math calculations for you. You will learn how to create formulas and functions to perform calculations in a spreadsheet.

Example formulas are: **=D15+D18+D21** : **=B4-B12** : **=A10/B15** : **=(B16+C16)*1.07** Do not use any spaces in formulas. Also, when creating formulas you may choose to either type the cell address or use the mouse to select the cell address.

Create Formula

You can create any type of math calculation on your own using the following mathematical operators:

Symbol	Meaning
=	equals **-** used to begin a calculation
+	addition
-	subtraction
*	multiplication
/	division
^	exponentiation
(	open parenthesis **-** used to begin a grouping
)	close parenthesis **-** used to close a grouping

The **numeric keypad** on the right side of the keyboard provides most of these operators. Excel follows the **mathematical order of hierarchy** where operators are processed in the order: negation, exponentiation, multiplication/division, and then addition/subtraction. Use parentheses to clarify the order of calculation in a formula.

Basic steps for creating a formula:

1. Click in the empty cell which will contain the formula.
2. Type an equal sign (=).
3. Type the cell address or click the cell that contains the first number.
4. Type the math operator **(+ - / * ^).**
5. Type the cell address or click the cell that contains the second number.
6. Continue in this manner until the formula is complete.
7. Use parenthesis for clarification.
8. Press the **Enter** key.

The following image depicts various formulas in an Excel spreadsheet which will be created in a following exercise:

	Budget	Actual	Variance
Office Supplies	300	226	=C20-D20
Office Postage	200	255	=C21-D21
Office Equipment & Furniture	800	679	=C22-D22
Miscellaneous Supplies	100	71	=C23-D23
Total:	**=C20+C21+C22+C23**	**=SUM(D20:D23)**	**=C24-D24**

AutoSum

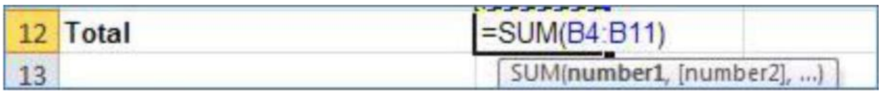
Adding is the most common math operation performed in Excel. The **Home** ribbon includes an **AutoSum** Σ button for adding. This button provides a shortcut to typing formulas.

Basic Steps for using AutoSum:

1. Move to the empty cell that will contain the formula.
2. Click on the **AutoSum** button.
3. **Proofread** the formula that Excel provides, make any necessary changes.

4. Press the **Enter** key or click the **check**  mark on the formula bar.

Click back on the **Budget sheet**  tab.



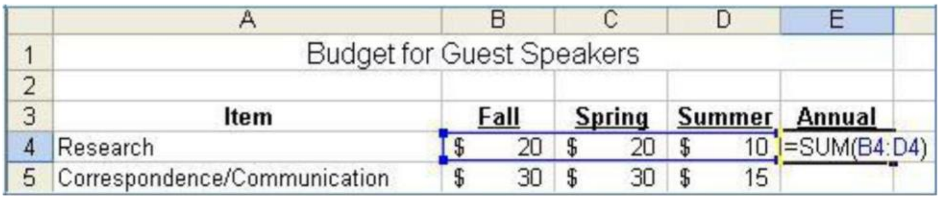
1. Select cell **B12**, click on the **AutoSum**  button, and then press the **Enter** key.
2. Repeat the **AutoSum** process for **cells C12, D12, E12**.

Note: You can copy formulas that refer to empty cells. After you type numbers in the empty cells, the formulas will be updated.

3. Click in cell **B4** and change the amount $20 to $50, and then press **Enter** key.

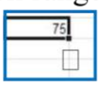
Note: Formula results are updated automatically in Excel. As you change any values that are referred to in a formula, the formula will reflect these changes.

4. Complete the AutoSum process for column E. Click in cell **E4**.



	A	B	C	D	E
1	Budget for Guest Speakers				
2					
3	Item	Fall	Spring	Summer	Annual
4	Research	$ 20	$ 20	$ 10	=SUM(B4:D4)
5	Correspondence/Communication	$ 30	$ 30	$ 15	

5. Click on the **AutoSum**  button to add the Research expenses for the three semesters.
6. Press the **Enter** key.
7. Select the cell range **B5:E5**, click the **AutoSum**  button, and then press **Enter** key.

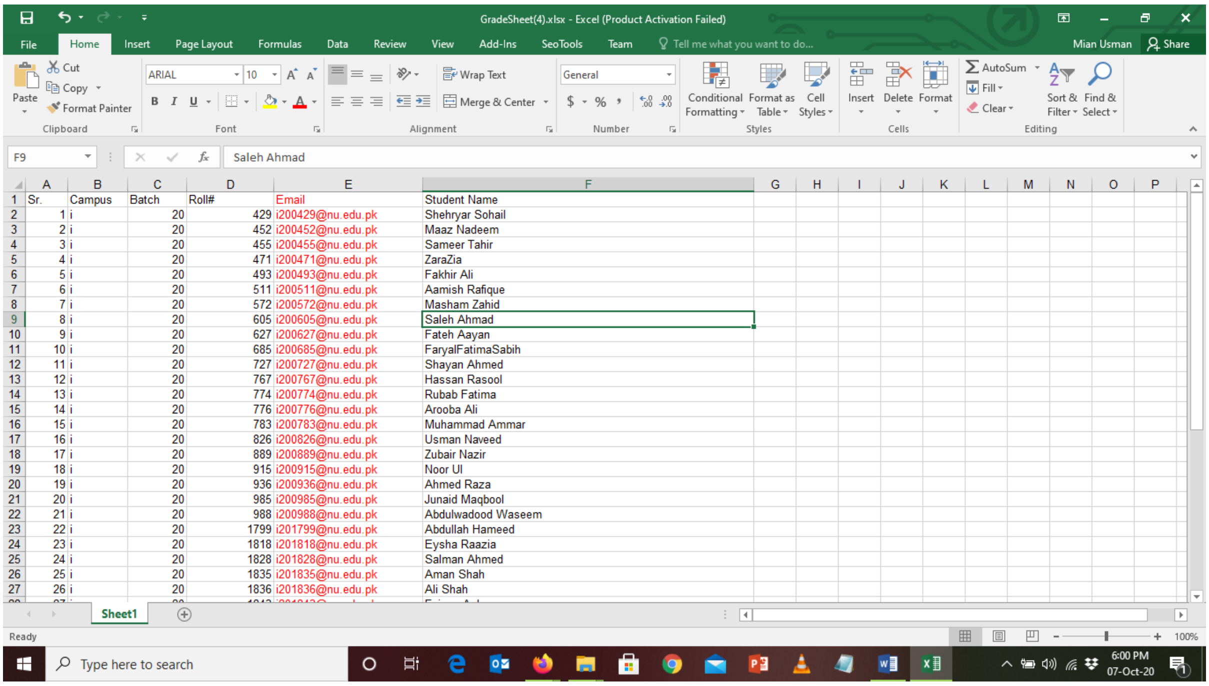
8. **Auto Fill**  this formula to the cell range **E6:E11**.

9. Copy this formula to the cell range E6:E11 by using the **Auto Fill** method illustrated above. Place the mouse pointer on the small solid square on lower right corner of cell E5, when the mouse pointer changes to a plus sign (Fill handle), then hold down on the right mouse button and drag the mouse down the designed cells (E6:E11) to copy the formula.

National University
of computer and emerging sciences

Tasks 1

You are provided with attendance sheet of a section and required to make a new sheet with following columns.



Sr.	Campus	Batch	Roll#	Email	Student Name
1	i	20	429	i200429@nu.edu.pk	Shehryar Sohail
2	i	20	452	i200452@nu.edu.pk	Maaz Nadeem
3	i	20	455	i200455@nu.edu.pk	Sameer Tahir
4	i	20	471	i200471@nu.edu.pk	ZaraZia
5	i	20	493	i200493@nu.edu.pk	Fakhir Ali
6	i	20	511	i200511@nu.edu.pk	Aamish Rafique
7	i	20	572	i200572@nu.edu.pk	Masham Zahid
8	i	20	605	i200605@nu.edu.pk	Saleh Ahmad
9	i	20	627	i200627@nu.edu.pk	Fateh Aayan
10	i	20	685	i200685@nu.edu.pk	FaryalFatimaSabih
11	i	20	727	i200727@nu.edu.pk	Shayan Ahmed
12	i	20	767	i200767@nu.edu.pk	Hassan Rasool
13	i	20	774	i200774@nu.edu.pk	Rubab Fatima
14	i	20	776	i200776@nu.edu.pk	Arooba Ali
15	i	20	783	i200783@nu.edu.pk	Muhammad Ammar
16	i	20	826	i200826@nu.edu.pk	Usman Naveed
17	i	20	889	i200889@nu.edu.pk	Zubair Nazir
18	i	20	915	i200915@nu.edu.pk	Noor Ul
19	i	20	936	i200936@nu.edu.pk	Ahmed Raza
20	i	20	985	i200985@nu.edu.pk	Junaid Maqbool
21	i	20	988	i200988@nu.edu.pk	Abdulwadood Waseem
22	i	20	1799	i201799@nu.edu.pk	Abdullah Hameed
23	i	20	1818	i201818@nu.edu.pk	Eysha Raazia
24	i	20	1828	i201828@nu.edu.pk	Salman Ahmed
25	i	20	1835	i201835@nu.edu.pk	Aman Shah
26	i	20	1836	i201836@nu.edu.pk	Ali Shah

Hint: you will split your roll no column and concatenate for formation of email address

Tasks 2

An Excel file of consumption of wine/beer in different countries is provided i.e drinks.xlsx. You are required to draw following graphs

1. Wine Serving on y-axis
   Countries on x-axis
2. Beer Serving on y-axis
   Countries on x-axis

From the following graphs you will report top consumer country of beer and wine.